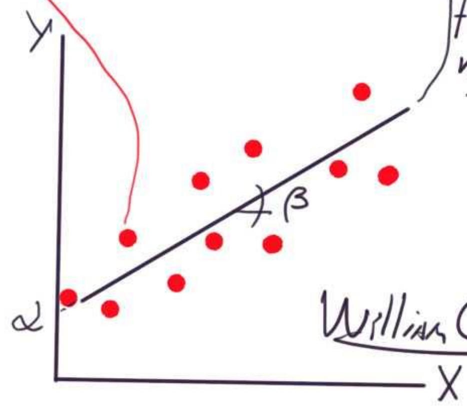
Prof. Luiz Paulo Lopes Fávero

Prints:

$Y_i = \alpha + \beta \cdot X_i + \mu_i$

$\hat{Y}_i = \alpha + \beta \cdot X_i$

fitted values



William Gosset

α , β

R^2

F (Fisher-Snedecor)

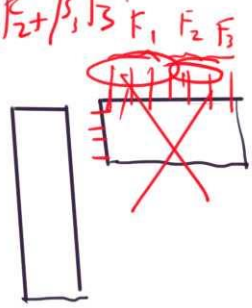
$$F = \frac{\frac{SQReg}{k}}{\frac{SQErros}{n-k-1}}$$

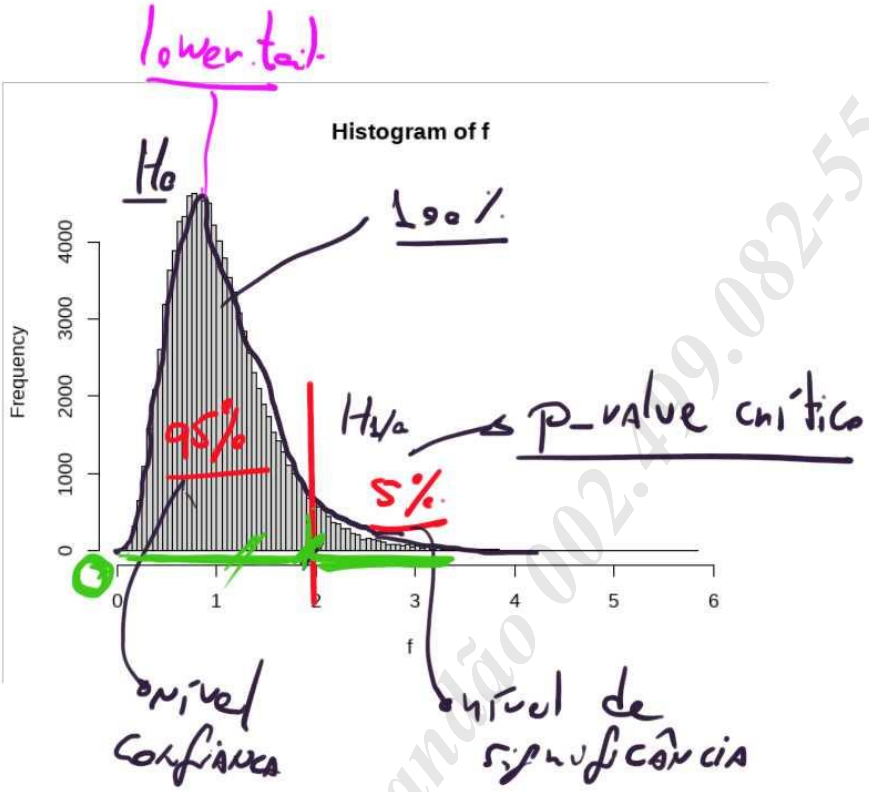
t-student

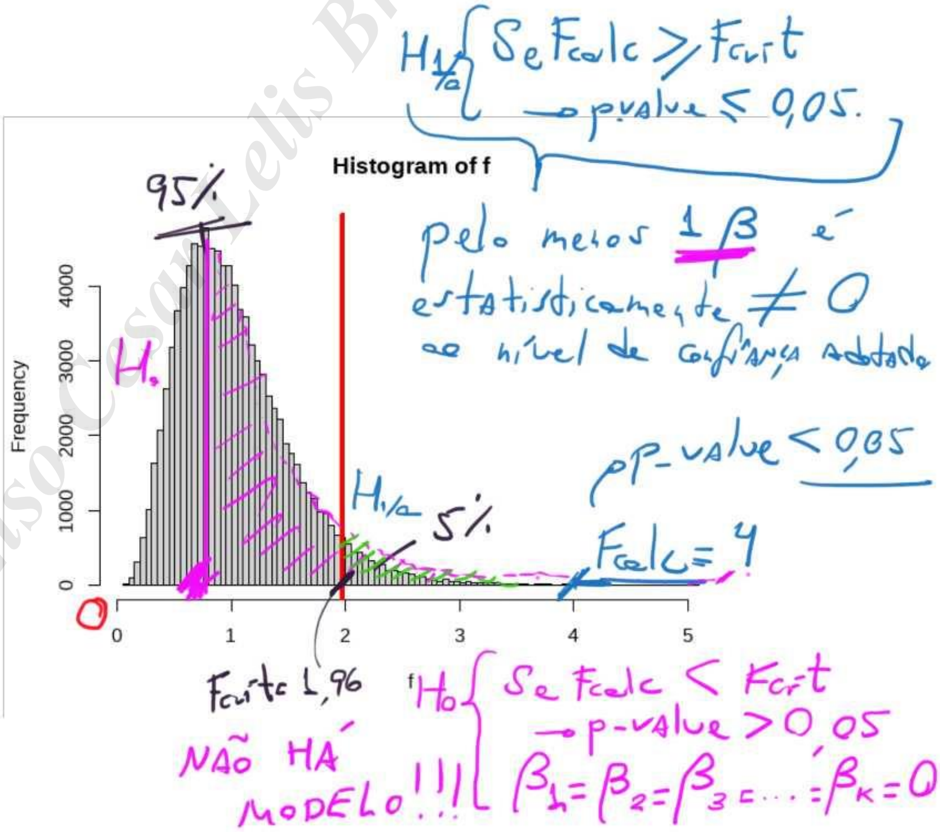
tidy $k > n$ (não estimável)

obs	Y	X_1	X_2	.....	$X_k = 31$
1					
⋮					
30					

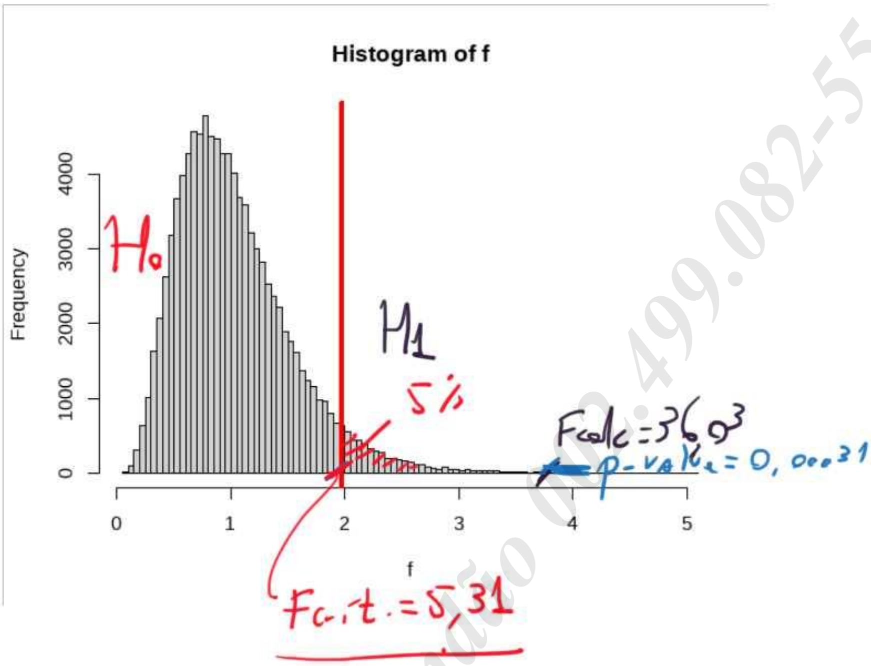
$Y = \alpha + \beta_1 \cdot F_1 + \beta_2 \cdot F_2 + \beta_3 \cdot F_3$

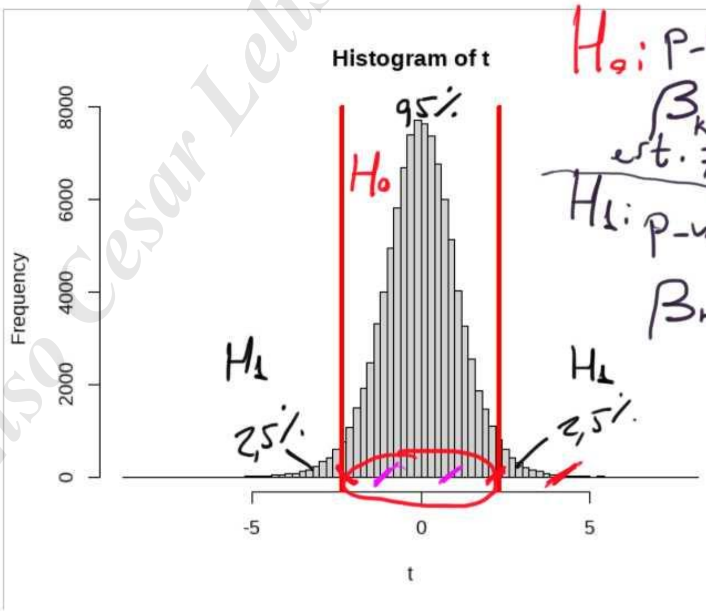



lower.tail
Histogram of f
H0
100%
Frequency
95%
H1/a
p_value crítico
5%
0
f
nível confiança
nível de significância


H1/a { Se Fcalc ≥ Fcrit
→ pvalue ≤ 0,05.
Histogram of f
95%
pelo menos 1 β é estatisticamente ≠ 0 ao nível de confiança adotado
H0
Frequency
p-value < 0,05
H1/a
5%
Fcalc = 4
0
Fcrit 1,96
f
H0 { Se Fcalc < Fcrit
→ p-value > 0,05
β1 = β2 = β3 = ... = βK = 0
NÃO HÁ MODELO!!!

Ex: Nosso Modelo:





H_0: p_value > 0,05
β_k não é est. $\neq 0$

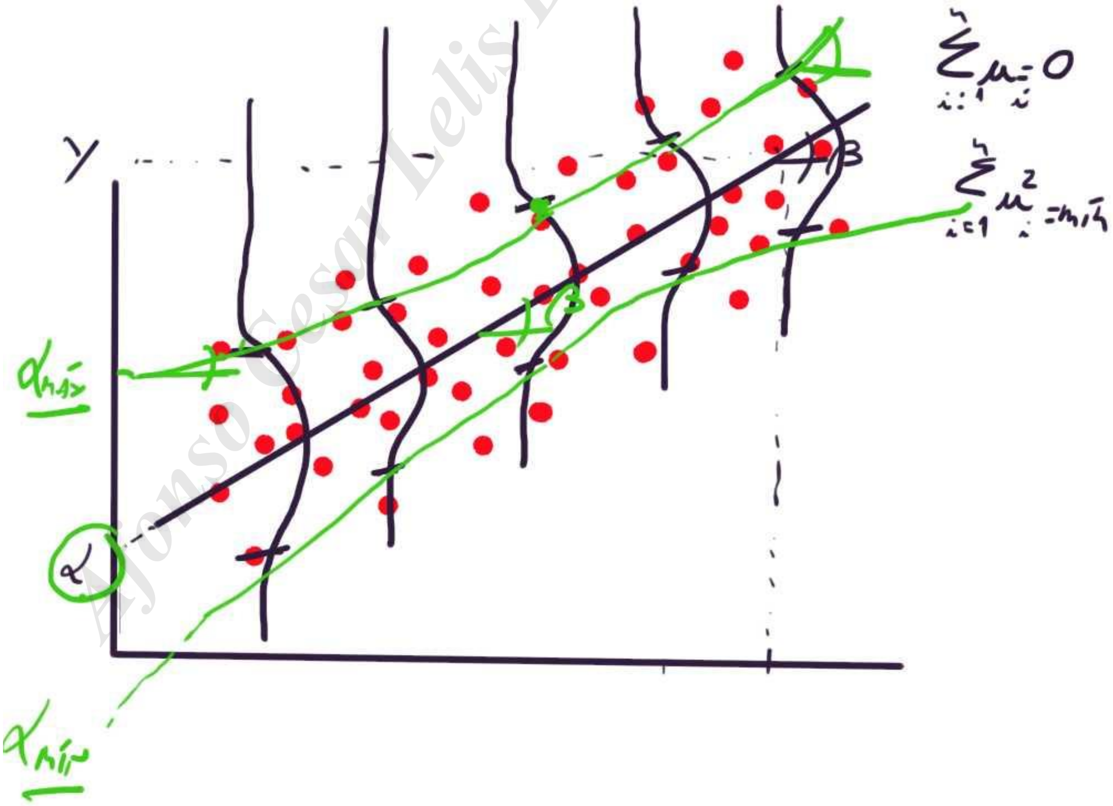
H_1: p_value $\leq$ 0,05
$\beta_k \neq 0$.

$$F = \frac{\dfrac{SQReg}{k \rightarrow df1}}{\dfrac{SQErros}{(n-k-1) \rightarrow df2}}$$

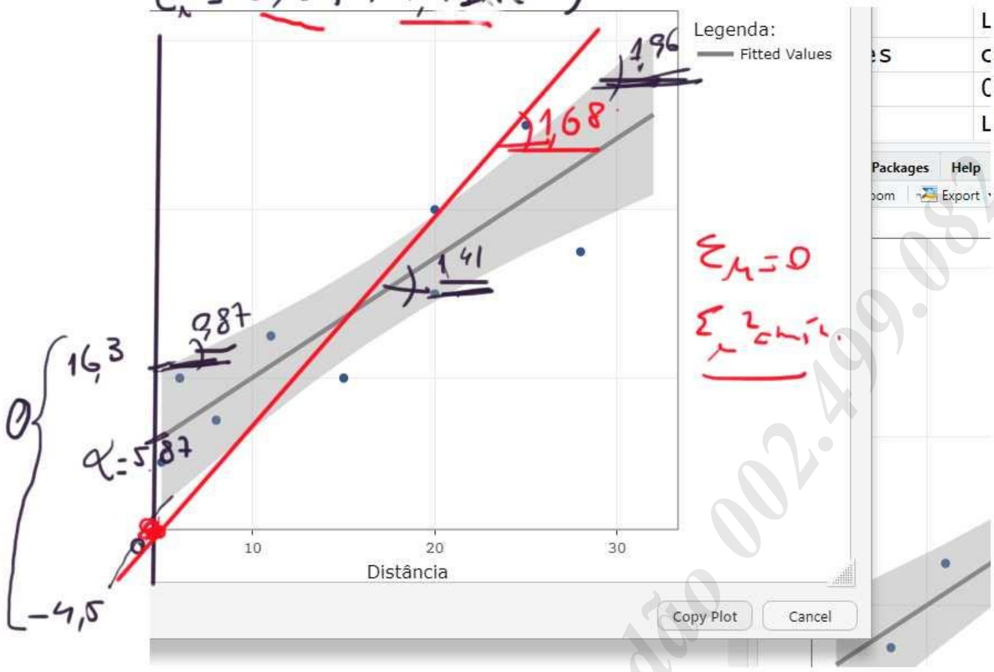
$10-1-1 = 8$

$df1 = 10$ (10 X's)

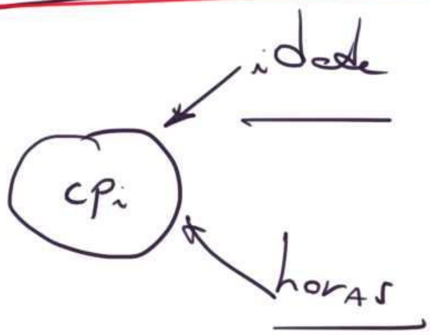
$df2 = 70$



$$\hat{t}_i = 5{,}87 + 1{,}41 \times (25)$$



$$\hat{cp}_i = \overset{11{,}97}{\alpha} + \overset{0{,}099}{\beta_1} \cdot idade_i + \overset{-0{,}40}{\beta_2} \cdot horas_i$$



Y quanti.
cpi

região

X quali.

Dummização: One-hot encoding

binárias
dicotômicas

País	cpi	região
Austrália	8,7	Oceania
Canadá	8,9	AN
EUA	7,1	AN
N. Zelândia	9,3	Oceania

AN: 8,0

Oceania: 9,0

	D	
Oceania	0	ref.
AN	1	altern.

α β

$\hat{cpi}_i = 9{,}0 - 1{,}0 \times AN$

$cpi_{oc} = 9{,}0 - 1{,}0 \times 0 = 9{,}0$

$cpi_{AN} = 9{,}0 - 1{,}0 \times 1 = 8{,}0$

	D
Oceania	1
AN	0

$\hat{cpi}_i = 8{,}0 + 1{,}0 \times Oceania$

$cpi_{oc} = 8{,}0 + 1{,}0 \times 1 = 9{,}0$

$cpi_{AN} = 8{,}0 + 1{,}0 \times 0 = 8{,}0$

País	cpi	região
AUS	8,7	OC
CAN	8,9	AN
EUA	7,1	AN
N.Zel	9,3	OC
Jordânia	2,0	Ásia

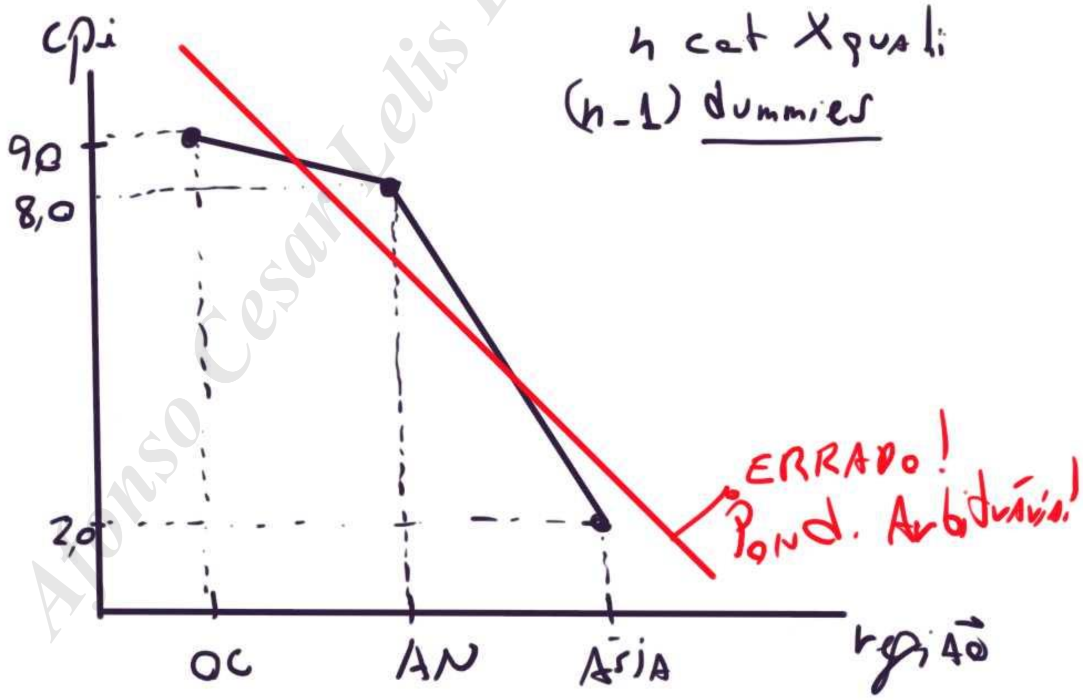
		D_1	D_2
ref.	OC	0	0
Alt1	AN	1	0
Alt2	ÁSIA	0	1

$$\hat{cpi} = \underbrace{9,0}_{\alpha} - \underbrace{1,0}_{\beta_1} \times \overbrace{D_1}^{AN} - \underbrace{7,0}_{\beta_2} \times \overbrace{D_2}^{ÁSIA}$$

$$\hat{cpi}_{oc} = 9,0 - 1,0 \times 0 - 7,0 \times 0 = 9,0$$

$$\hat{cpi}_{AN} = 9,0 - 1,0 \times 1 - 7,0 \times 0 = 8,0$$

$$\hat{cpi}_{AS} = 9,0 - 1,0 \times 0 - 7,0 \times 1 = 2,0$$



n cat X quali
(n-1) dummies